

packets. It also allows the administrator to select the interface to capture packets on. The 'any' option in the interface list matches all of the network interfaces.

Once packet capturing has been stopped, the captured network packets can be written to a file for sharing or later analysis. The *File>Save>...* or *File > Save as...* menu item allows the user to specify the files to save the packet into. Wireshark supports a variety of file formats.

9. Kdump

RHEL offers the Kdump software for the capture of kernel crash dumps. This software works by using the *kexec* utility on a running system to boot a secondary Linux kernel without going through a system reset. The Kdump software is installed by default in RHEL 7 through the installation of the *kexec-tools* package. The package provides the files and command line utilities necessary for administering Kdump from the command line.

```
# yum -y install kexec-tools system-config-kdump.
```

The Kdump crash dump mechanism is provided through the *kdump* service. Administrators interested in enabling the collection of kernel crash dumps on their systems must ensure that the *kdump* service is enabled and started on each system.

```
# systemctl enable kdump.
# systemctl start kdump.
```

With the *kdump* service enabled and started, kernel crash dumps will begin to be generated during system hangs and crashes. The behaviour of the kernel crash dump and collection can be modified in various ways by using the */etc/kdump.conf* configuration file.

By default, Kdump captures crash dumps locally to *crash*

dump files located in subdirectories under the */var/crash* path.

10. SystemTap

The SystemTap framework allows easy probing and instrumentation of almost any component within the kernel. It provides administrators with a flexible scripting language and library by leveraging the kprobes facility within the Linux kernel. Using kprobes, kernel programmers can attach instrumentation code to the beginning or end of any kernel function. SystemTap scripts specify where to attach probes and what data to collect when the probe executes.

SystemTap requires symbolic naming for instructions within the kernel. So it depends on the following packages, which are not usually found on production systems. These packages will pull in any required dependencies. The packages are:

- 1) Kernel-debuginfo
- 2) Kernel-devel
- 3) Systemtap

Using stap to run SystemTap scripts: The SystemTap package provides a variety of sample scripts that administrators may find useful for gathering data on their systems. The scripts are stored in */usr/share/doc/systemtap-client-*/examples*. These scripts are further divided into several different subdirectories based on what type of information they have been asked to collect. SystemTap scripts have an extension of *.stp*.

To compile and run these example scripts, or any other SystemTap script for that matter, administrators use the *stap* command. 

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